

HEALTHCONNECT CLINIC

ANALYTICS REPORT

Project focus: Understanding appointment attendance and no-show patterns to support evidence-based operational decisions.

Executive Summary

Week 5 moved the HealthConnect Data Analytics track from the planning and data understanding stage established in Week 4 into practical exploratory analysis. The analysis used the appointment level dataset to calculate the selected KPIs and compare no-show behaviour across previous no-show history, booking lead time, appointment type, appointment day, age group, reminder channel, distance to clinic and waiting time.

The dashboard indicates that no-shows represent the largest appointment outcome category, at 48.5% of appointments, compared with an attendance rate of 46.3% and a cancellation rate of 5.3%. The strongest visible patterns are associated with previous no-show history and booking lead time. Higher previous no-show counts correspond with higher no-show rates, while appointments booked further in advance show progressively higher no-show rates.

Objectives

- Prepare and validate the relevant appointment data without overwriting the original dataset.
- Calculate and interpret 3-5 KPIs proposed in Week 4.
- Explore attendance and no-show patterns across relevant appointments and patient characteristics.
- Develop an analytical dashboard containing evidence based visualizations.
- Translate the strongest patterns into business insights and practical recommendations.
- Document limitations and distinguish association from causation.

Data Preparation & Quality Summary

The Week 4 assessment identified appointment_id as the appointment level identifier and found no duplicate appointment IDs. Approximately 1% of values were empty in distance_to_clinic_km and waiting_time_minutes. These fields were therefore treated as variables requiring caution during interpretation rather than assuming missing values represented zero.

- Primary appointment identifier: appointment_id.
- Duplicate appointment IDs: none identified in the Week 4 check.
- Missingness: approximately 1% in distance_to_clinic_km and waiting_time_minutes.
- Key categorical/analytical fields include appointment outcome, appointment type, appointment day, reminder channel, age group and booking lead group.

- The original dataset should remain preserved separately from any transformed/processed version.

KPI Development & Results

KPI	Dashboard Result	Measurement Focus	Business Value
No-Show Rate	48.5%	No-show appointments ÷ relevant scheduled appointments	Shows the scale of missed appointments.
Attendance Rate	46.3%	Attended appointments ÷ relevant scheduled appointments	Shows the overall successful attendance level.
Cancellation Rate	5.3%	Cancelled appointments ÷ relevant scheduled appointments	Shows the proportion of appointments ending in cancellation.
Reminder-Associated Attendance Rate	46.3%	Attendance compared across reminder channels	Supports investigation of reminder-related attendance patterns.
Repeat No-Show Rate	55.4%	No-show appointments involving patients with previous no-show history	Highlights recurring no-show behaviour.

Exploratory Data Analysis

Appointment Outcome Distribution

The dashboard shows 5,000 total appointments. No-shows account for approximately 48.5% (about 2,423 appointments), attendance for approximately 46.3%, and cancellations for approximately 5.3%. This establishes no-shows as the dominant outcome category and the central operational issue for further investigation.

Previous No-Show History

No-show rates rise as previous no-show history increases. The dashboard shows approximately 43.3% for patients with no previous no-shows, 59.4% for one previous no-show, 67.9% for two, 66.7% for three, 67.8% for four and 100% for five previous no-shows. The 100% value should be interpreted carefully because the underlying group may be small.

Booking Lead Time

The booking lead time analysis shows a clear upward pattern: approximately 27.8% for 0-7 days, 33.6% for 8-14 days, 43.2% for 15-30 days and 60.5% for 31+ days. This suggests that appointments scheduled substantially in advance are associated with higher no-show rates.

Reminder Channel

Attendance varies by reminder channel. The dashboard shows approximately 49.6% attendance for SMS, 46.5% for Email, 44.6% for WhatsApp and 42.7% where no reminder was recorded. SMS has the highest observed attendance among the displayed channels, while the no-reminder group has the lowest. These results indicate association, not proof that a particular channel causes attendance.

Appointment Type

No-show rates are highest for Follow-up appointments at approximately 51.2%, followed by Diagnostic Test at 49.7%, Specialist Consultation at 47.4% and General Consultation at 46.6%.

Appointment Day

No-show rates vary across appointment days, with the dashboard showing the highest values around Sunday/Saturday and lower values around Friday/Tuesday. The differences are useful for operational monitoring, although they should be assessed alongside appointment volumes before major scheduling changes are made.

Age Group

The age-group comparison shows relatively high no-show rates among 55-64, 25-34 and 18-24 groups (around 50%), while the 65+ group is lower at about 45.1%. The differences are comparatively modest, so age should not be treated as the strongest driver based on this dashboard alone.

Distance to Clinic

No-show rates are approximately 51.4% for 0-2 km, 45.3% for 2.5-5 km, 46.5% for 5-10 km and 51.0% for 10+ km. The relationship is not linear, so distance alone does not appear to provide a simple explanation for no-shows.

Waiting Time

No-show rates are approximately 47.8% for 0-5 minutes, 49.2% for 16-30 minutes, 47.6% for 31-60 minutes and 66.7% for 60+ minutes. The 60+ minute group stands out and warrants further investigation, while the limited missing values in `waiting_time_minutes` should be considered.

Business Insights

1. No-shows are the largest appointment outcome.

With a no-show rate of 48.5% versus an attendance rate of 46.3%, nearly half of recorded appointments end in a no-show. This represents a significant appointment capacity and scheduling concern for HealthConnect.

2. Previous no-show behaviour is the strongest visible segmentation pattern.

No-show rates increase substantially from 43.3% among patients with no previous no-shows to roughly 59-68% among patients with multiple previous no-shows. This identifies repeat no-show history as an important risk segment.

3. Longer booking lead times are associated with higher no-show rates.

No-show rates rise from 27.8% for appointments booked 0-7 days ahead to 60.5% for appointments booked 31+ days ahead. This suggests that long gaps between booking and appointment may be an important operational pattern to investigate.

4. Reminder channel is associated with different attendance levels.

SMS has the highest displayed attendance rate at 49.6%, while the no-reminder group has the lowest at 42.7%. This supports further evaluation of reminder timing and channel effectiveness.

5. Waiting times above 60 minutes are a notable risk signal.

The 60+ minute waiting time group has a no-show rate of approximately 66.7%, considerably higher than the other waiting time groups. This may justify closer operational review of appointments associated with long expected waits.

Business Recommendations

1. Introduce a targeted engagement workflow for patients with previous no-show history. Consider stronger reminders, confirmation requests and follow-up procedures for repeat no-show segments.
2. Review appointments with 31+ days of booking lead time. Where clinically and operationally appropriate, consider confirmation closer to the appointment date and evaluate whether some appointment slots can be scheduled nearer to the required date.
3. Evaluate SMS as a priority reminder channel because it has the highest observed attendance rate in the dashboard. A controlled comparison should be used before concluding that SMS itself causes higher attendance.
4. Investigate the operational causes behind the 60+ minute waiting-time group. Review whether these appointments are linked to a particular appointment type, days or clinic processes.
5. Monitor no-show performance using the dashboard KPIs on a recurring basis, with segmentation by booking lead time, previous no-show history and reminder channel so that changes can be identified early.

Limitations

- The dataset is fictional and anonymised; findings are intended for the HealthConnect Experience Lab and should not be treated as direct clinical evidence.
- Approximately 1% of values in distance_to_clinic_km and waiting_time_minutes are missing, which can affect variable specific comparisons.
- The analysis is primarily descriptive. Observed relationships are associations and do not establish causation.
- Some extreme segment results, such as the 100% no-show rate for five previous no-shows, may be based on a small number of records and should be validated with group counts.
- The dataset may omit behavioural, socioeconomic, operational and other contextual factors that influence appointment attendance.
- Incorrect treatment of cancellations, missing values or category labels could affect KPI calculations and comparisons.

Week 5 Conclusion

The Week 5 analysis successfully moved the HealthConnect project from initial planning into evidence generation. The dashboard provides a baseline view of appointment outcomes and highlights several segments that warrant deeper investigation. The strongest observed patterns relate to previous no-show history and booking lead time, while reminder channel and long waiting time also provide useful operational signals.